

Fig. 4: An actual-size, single-side PCB for the ASK communication trainer

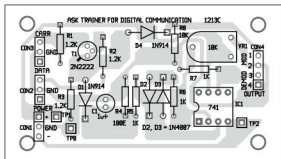


Fig. 5: Component layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - 741 operational amplifier
- T1 - 2N2222 npn transistor
- D1, D4 - 1N914/1N4148 diode
- D2, D3 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R3 - 1.2-kilo-ohm
- R4 - 150-ohm
- R5-R7 - 1-kilo-ohm
- R8 - 10-kilo-ohm
- VR1 - 4.7-kilo-ohm preset

Capacitors:

- C1 - 1 μ F, 25V electrolytic

Miscellaneous:

- CON1-CON3 - 3-pin connector
- CON4 - 4-pin connector

few other components.

The circuit comprises modulator and demodulator sections. The modulator is built around transistor T1. Digital modulating/baseband signal (such as, 5V, 500Hz square wave) is given to the base of transistor T1. You can use 555 timer in astable mode or a signal generator to generate this signal and feed to the circuit through CON2. Similarly, 5V, 10kHz signal generated by the function generator is fed to CON3 as a carrier signal, which is actually connected to the collector of transistor T1.

The ASK modulated output is produced at the emitter of transistor T1 corresponding to the input data square wave by switching transistor T1. The modulated output signal is available across pins 1 and 2 of output connector CON4. You can observe this signal using an oscilloscope.

The demodulation section has two segments: an envelope detector built around diode D1, resistor R4 and capacitor C1, and a comparator built around operational amplifier 741 (IC1) along with protection diodes D2 and D3. Envelope detector is nothing but a combination of half-wave rectifier built with diode

D1 and an RC network with time constant $T = R4 \times C1$.

The produced amplitude or envelope of the envelope detector is fed to non-inverting pin of comparator IC1, which compares it with the reference voltage selected through preset VR1 at the inverting pin. The final demodulated output is available across pins 3 and 4 of output connector CON4. You can observe this demodulated signal well with an oscilloscope; it should match the data at input connector CON2.

Construction and testing

An actual-size, single-side PCB for the ASK communication trainer is shown in Fig. 4 and its component layout in Fig. 5. After assembling the circuit on PCB, enclose it in a suitable plastic case.

Fix CON1 through CON4 on the front side of panel for power supply, input data signal, carrier signal and modulated and demodulated signal, respectively. The connectors also serve as test points. Refer the test points table and check the signals using an oscilloscope. ●

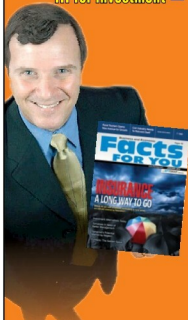
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